

A.3 Work, energy and power

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

Take $g = 9.81 \text{ m s}^{-2}$. Show your working.

Question 1 — The café tray

(5 marks)

A waiter carries a loaded tray, weight 30 N, across a level café floor at constant velocity.

- (a) Define the work done by a force, including the role of the angle between force and displacement.
[2]
- (b) Explain why, during the level crossing, the waiter's upward supporting force does *no* work on the tray.
[2]
- (c) State one force in this scene that *does* do work on the tray, however small.
[1]

Question 2 — The e-scooter commute

(7 marks)

An e-scooter and rider (combined mass 90 kg) climb a hill of vertical height 15 m along a 120 m slope, at a steady 5.0 m s^{-1} .

- (a) Calculate the time taken for the climb.
[1]
- (b) Calculate the useful power the motor must deliver against gravity.
[3]
- (c) The motor is 70% efficient. Calculate the electrical power it draws from the battery, and state what happens to the rest.
[3]

Question 3 — Archery club

(7 marks)

A training bow behaves like a spring of constant $k = 400 \text{ N m}^{-1}$. An archer draws the string back 0.60 m and shoots an arrow of mass 40 g.

- (a) Calculate the elastic potential energy stored at full draw.
[2]
- (b) Assuming all of it transfers to the arrow, calculate the arrow's launch speed.
[3]
- (c) A clubmate reckons that drawing only half as far would store half the energy. Show quantitatively what fraction it actually stores.
[2]

Question 4 — The show-off weightlifter

(7 marks)

A weightlifter raises a 120 kg barbell through 2.0 m at constant (slow) speed, holds it overhead, then lowers it back down.

- (a) Calculate the work done on the barbell by the lifter during the lift.
[2]
- (b) The barbell's kinetic energy is unchanged by the lift. Reconcile this with your answer to (a): where did the energy go?
[2]
- (c) While the barbell is held stationary overhead the lifter trembles with effort. State how much work is done on the barbell during the hold, and why the lifter tires anyway.
[3]

Answers

Question 1 — The café tray

- (a) $W = Fs \cos \theta$: the force, times the displacement, times the cosine of the angle between them.
- (b) The supporting force is vertical while the displacement is horizontal, so $\theta = 90^\circ$, $\cos \theta = 0$ and therefore $W = 0$.
- (c) Air resistance does negative work; so does friction at the hand if the tray shifts. Gravity does no work here, being perpendicular to the motion.

Question 2 — The e-scooter commute

- (a) $t = 120/5.0 = 24 \text{ s}$.
- (b) $P = \frac{mgh}{t} = \frac{90 \times 9.81 \times 15}{24} = 550 \text{ W}$.
- (c) $P_{\text{in}} = 550/0.70 = 790 \text{ W}$. The remaining $\approx 240 \text{ W}$ becomes internal energy, heating the motor and electronics.

Question 3 — Archery club

- (a) $E = \frac{1}{2}kx^2 = \frac{1}{2}(400)(0.60)^2 = 72 \text{ J}$.
- (b) $\frac{1}{2}mv^2 = 72 \Rightarrow v = \sqrt{2 \times 72/0.040} = 60 \text{ m s}^{-1}$.
- (c) Since $E \propto x^2$, half the draw stores one *quarter* of the energy, 18 J, not half.

Question 4 — The show-off weightlifter

- (a) $W = mgh = 120 \times 9.81 \times 2.0 = 2.4 \text{ kJ}$ (2350 J).
- (b) It became gravitational potential energy of the barbell. The work–energy theorem applies to the *net* work, which is zero here: the lifter’s positive work plus gravity’s equal negative work.
- (c) Zero work on the barbell, because there is no displacement. The lifter tires because muscle fibres consume chemical energy to maintain tension, none of which is transferred to the barbell.